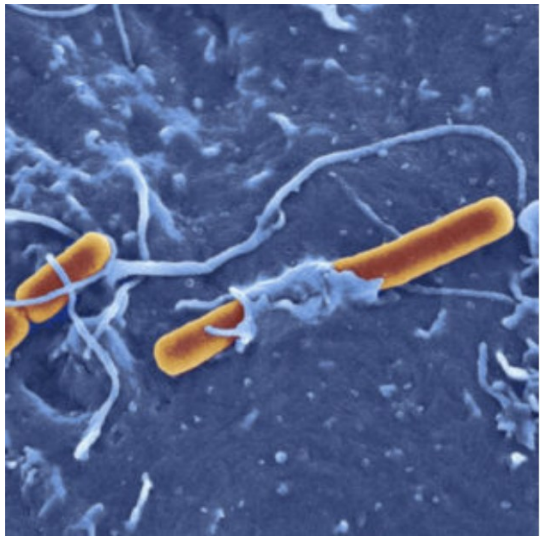




Genomic & Epidemiological Surveillance of Bacterial Pathogens

22 – 26 June 2026

Module 5: Genome Assembly and Annotation

Part 2 – Genome Annotation



Trinity College Dublin
Coláiste na Tríonóide, Baile Átha Cliath
The University of Dublin

Bacterial Genomic Surveillance - Ireland and Northern Ireland

Course Manual Dublin 2026

- Genomic Capacity Development for Public Health Settings
- World Café: Translating Genomics into Public Health Action
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- [Module 6: Web tools for genomic epidemiology](#)
- Group Task
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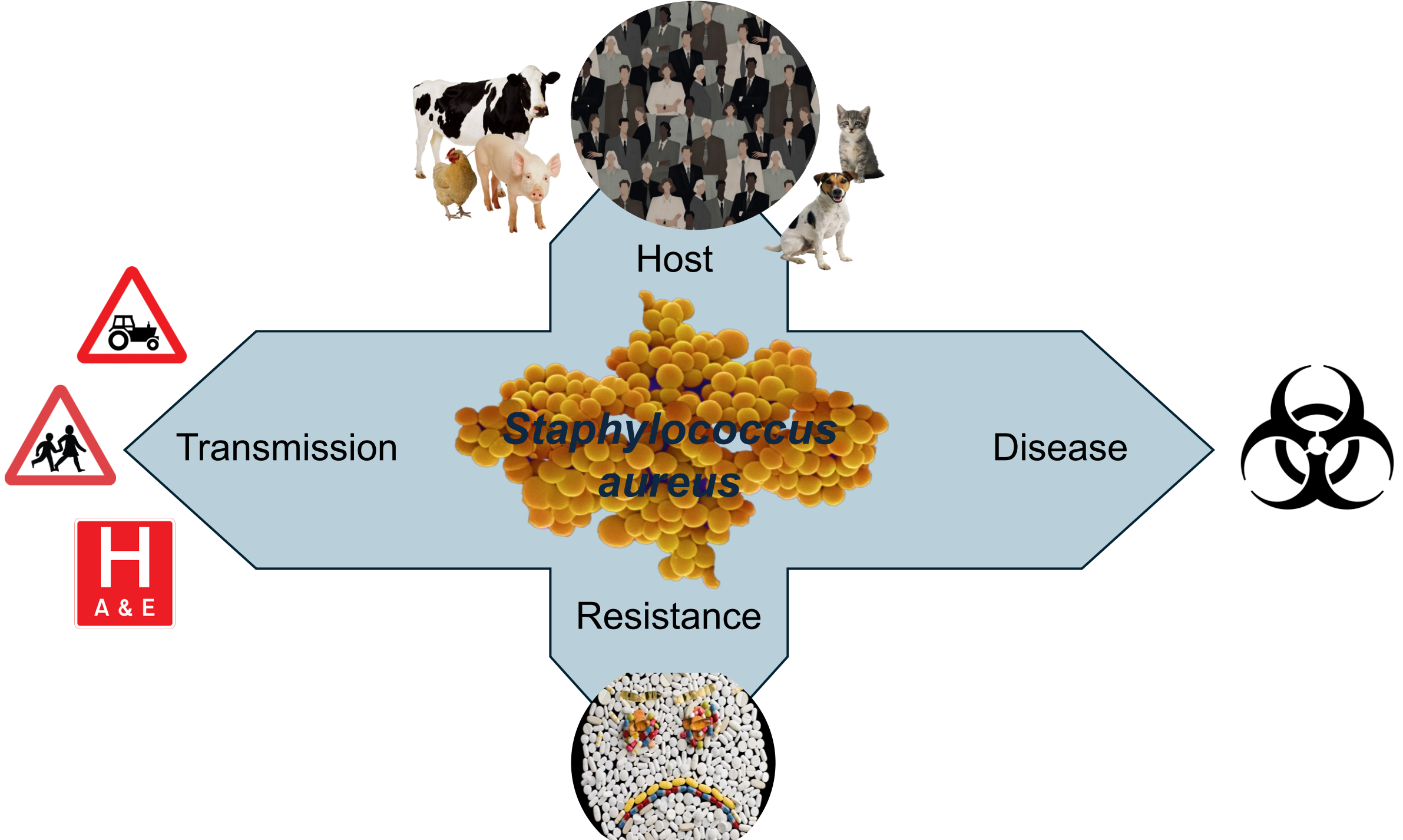
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Genome Assembly and Analysis - Dublin 2026

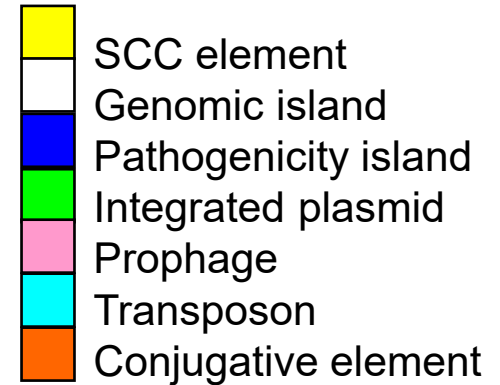
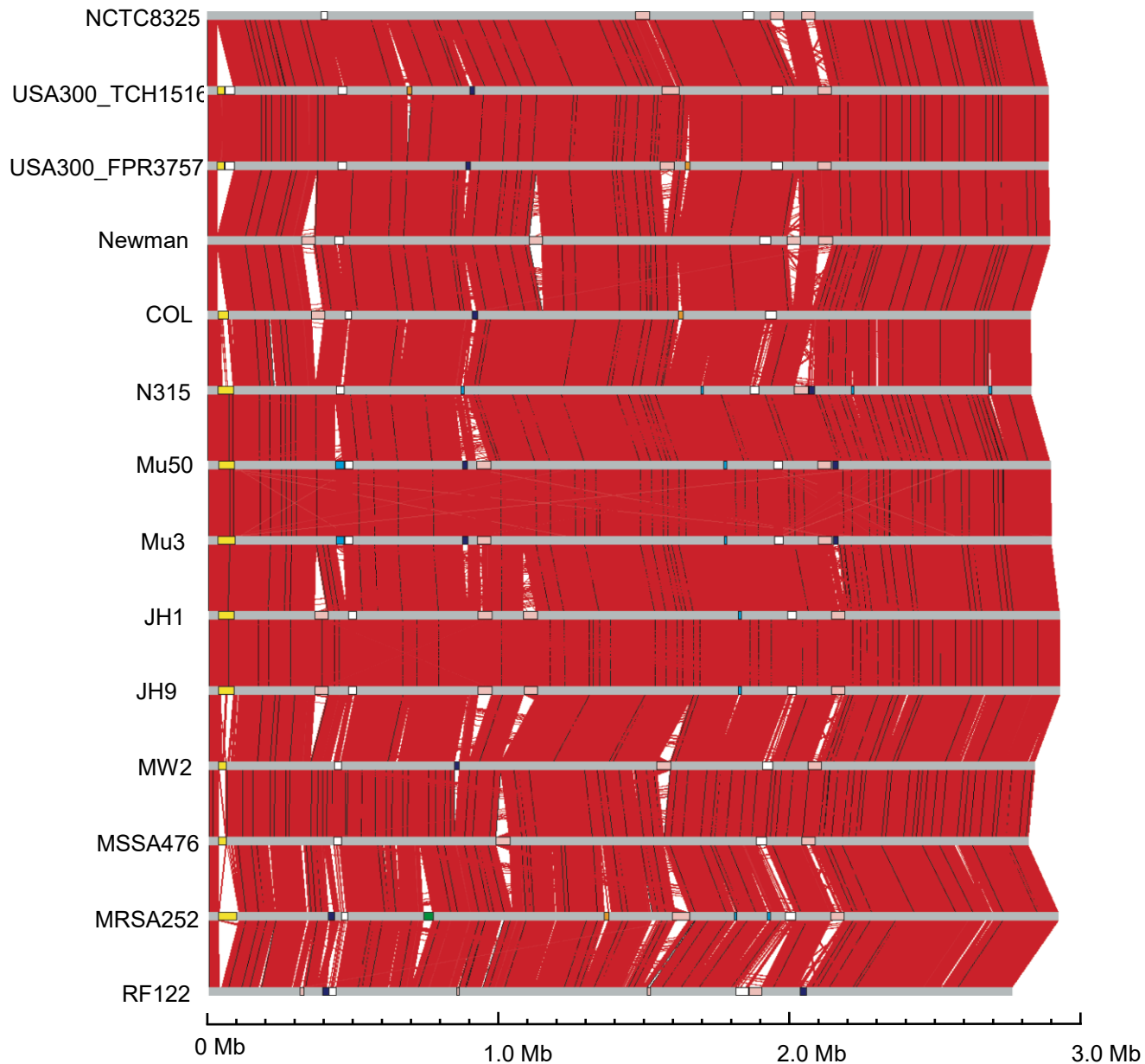
Module Leads: César Rodríguez and Tomás Caride Poklepovich

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S aureus core and accessory genome



Whole genome sequencing of a MRSA outbreak

The NEW ENGLAND JOURNAL of MEDICINE

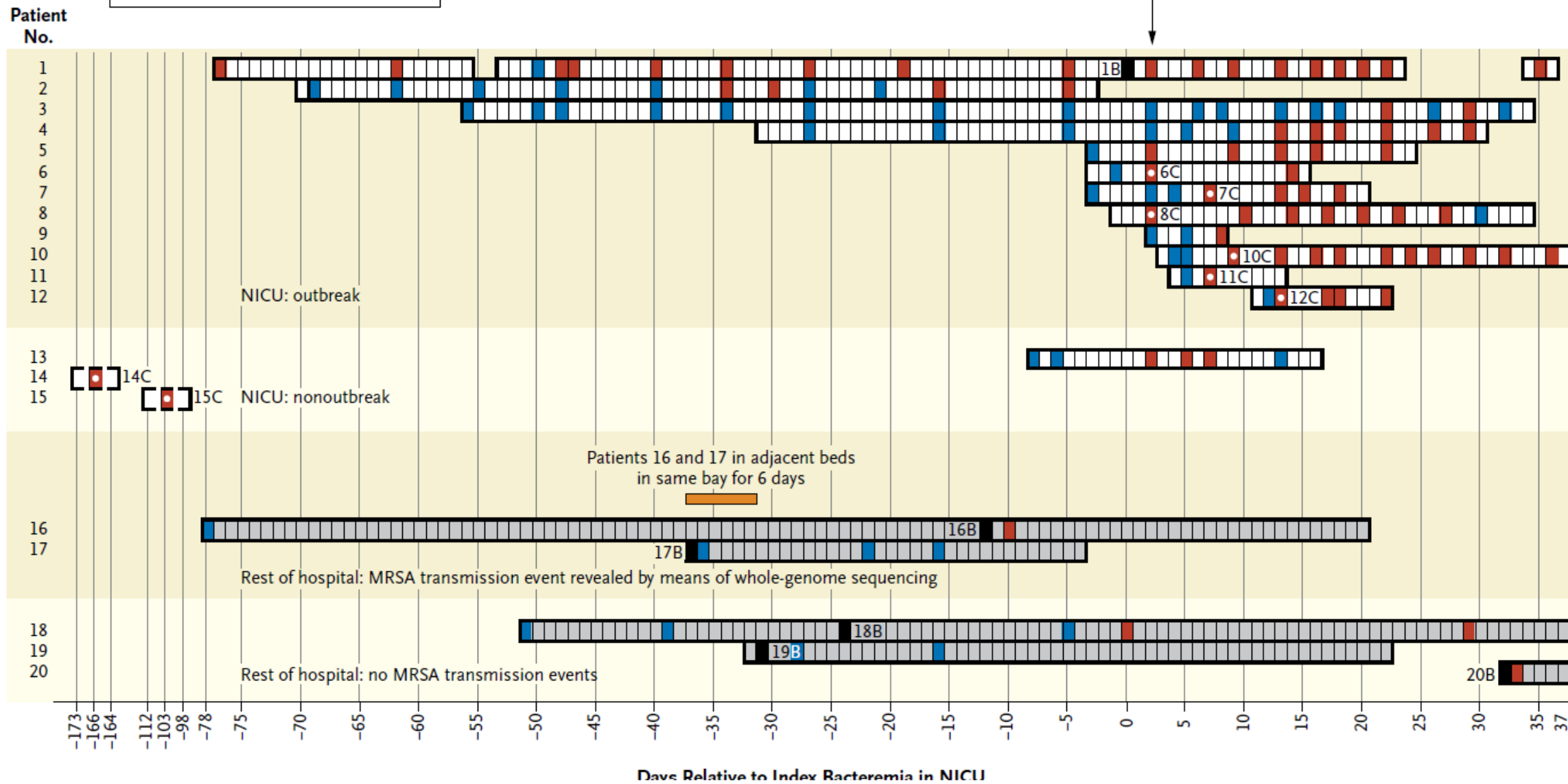
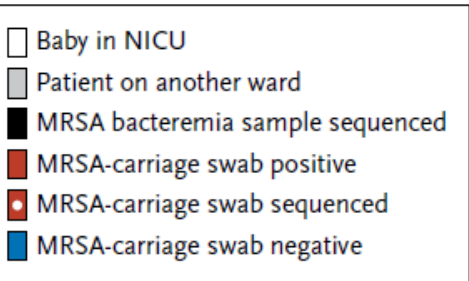
ORIGINAL ARTICLE

Rapid Whole-Genome Sequencing for Investigation of a Neonatal MRSA Outbreak

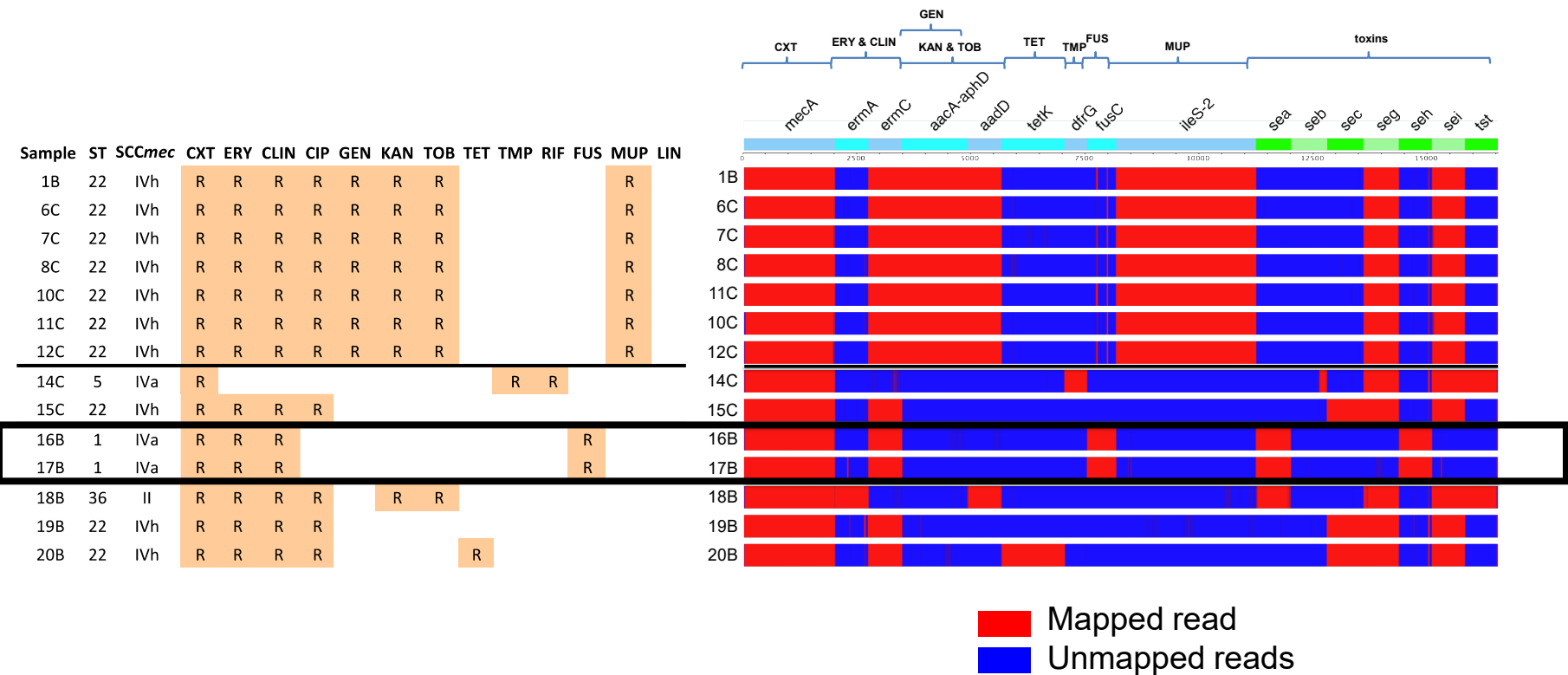
Claudio U. Köser, B.A., Matthew T.G. Holden, Ph.D., Matthew J. Ellington, D.Phil., Edward J.P. Cartwright, M.B., B.S., Nicholas M. Brown, M.D., Amanda L. Ogilvy-Stuart, F.R.C.P., Li Yang Hsu, M.R.C.P., Claire Chewapreecha, B.A., Nicholas J. Croucher, M.A., Simon R. Harris, Ph.D., Mandy Sanders, B.Sc., Mark C. Enright, Ph.D., Gordon Dougan, Ph.D., Stephen D. Bentley, Ph.D., Julian Parkhill, Ph.D., Louise J. Fraser, Ph.D., Jason R. Betley, Ph.D., Ole B. Schulz-Trieglaff, Ph.D., Geoffrey P. Smith, Ph.D., and Sharon J. Peacock, Ph.D., F.R.C.P.

NEJM (2012) 366:2267-75

- Two MRSA isolates included in the analysis (16B & 17B) were from patients on different wards in a hospital, and not thought to be part of the outbreak.
- So is it community transmitted or it is transmitted inside the hospital?



Congruence of antibiotic resistance phenotype and genotype



Antibiotics tested:

cefoxitin (CXT)
erythromycin (ERY)
ciprofloxacin (CIP)
gentamicin (GEN)
tetracycline (TET)
rifampicin (RIF)
fusidic acid (FUS)
mupirocin (MUP)

Disk susceptibility testing according to BSAC criteria on ISO sensitest agar

clindamycin (CLIN)*
kanamycin (KAN)*
tobramycin (TOB)*
trimethoprim (TMP)*
linezolid (LIN)*

* Vitek 2 system (bioMérieux) according to EUCAST 2008 guidelines

The module exercise

We can compare 16B to the genomes of ST1 community isolates in public databases:

- MSSA476 (UK)
- MW2 (USA)

The more recently isolated 16B is more drug resistant than either MSSA476 & MW2 (both older and community isolates):

- 16B - penicillin^R, fusidic acid^R, methicillin^R, erythromycin^R
- MSSA476 – penicillin^R, fusidic acid^R
- MW2 – penicillin^R, methicillin^R

Using the 16B raw sequence data you will predict the resistome, assemble, annotate and use comparative genomics to investigate the genetic basis of the antibiotic resistance of 16B and how this evolved

Aims of this exercise

- Use a resistome prediction tool to **identify the genetic determinants for antibiotic resistance** from sequencing reads
- To practice how **short-read** data can be **assembled into a draft genome**
- **Order the draft genome** against a reference sequence
- **Annotate** the reordered draft genome
- To show how, using comparative genomics, **regions of difference that distinguish genomes** can be identified and analysed
- Identify **the genetic basis of resistance** and be able to **explain the evolution of resistance** in the isolates investigated

Module structure

Identification of antibiotic resistance genes in 16B, MW2 and MSSA476 - **ARIBA**

de novo assembly of 16B - **UNICYCLER**

Calculate assembly statistics - **QUAST**

Order assembly against MSSA476 - **ABACAS**

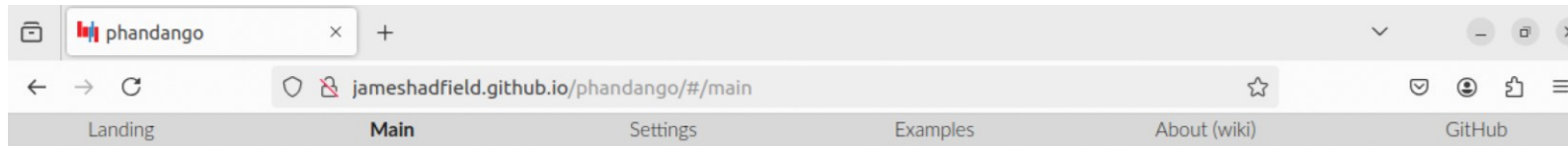
Comparative analysis against MSSA476 - **BLASTN** and **ACT**

Check for mis-assemblies and ordering errors – **SNIPPY** and **ACT**

Automated annotation – **BAKTA** and **ACT**

Comparative analysis against MW2 and MSSA476 - **BLASTN** and **ACT**

Comparing the resistome



Draft genome assembly

QUAST

Quality Assessment Tool for Genome Assemblies by [CAB](#)

22 July 2024, Monday, 19:02:11

[View in Icarus contig browser](#)

All statistics are based on contigs of size ≥ 500 bp, unless otherwise noted

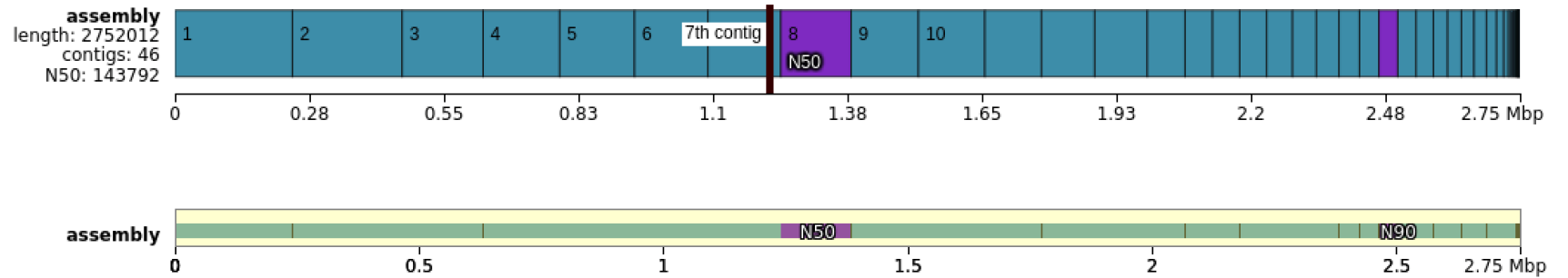
Statistics without reference ☐ assembly

# contigs	46
# contigs (≥ 0 bp)	78
# contigs (≥ 1000 bp)	38
# contigs (≥ 5000 bp)	33
# contigs (≥ 10000 bp)	31
# contigs (≥ 25000 bp)	29
# contigs (≥ 50000 bp)	18
Largest contig	240 345
Total length	2 752 012
Total length (≥ 0 bp)	2 757 710
Total length (≥ 1000 bp)	2 745 934
Total length (≥ 5000 bp)	2 733 517
Total length (≥ 10000 bp)	2 719 234
Total length (≥ 25000 bp)	2 683 718
Total length (≥ 50000 bp)	2 286 822
N50	143 792
N90	38 605
auN	129 830
L50	8
L90	23
GC (%)	32.6

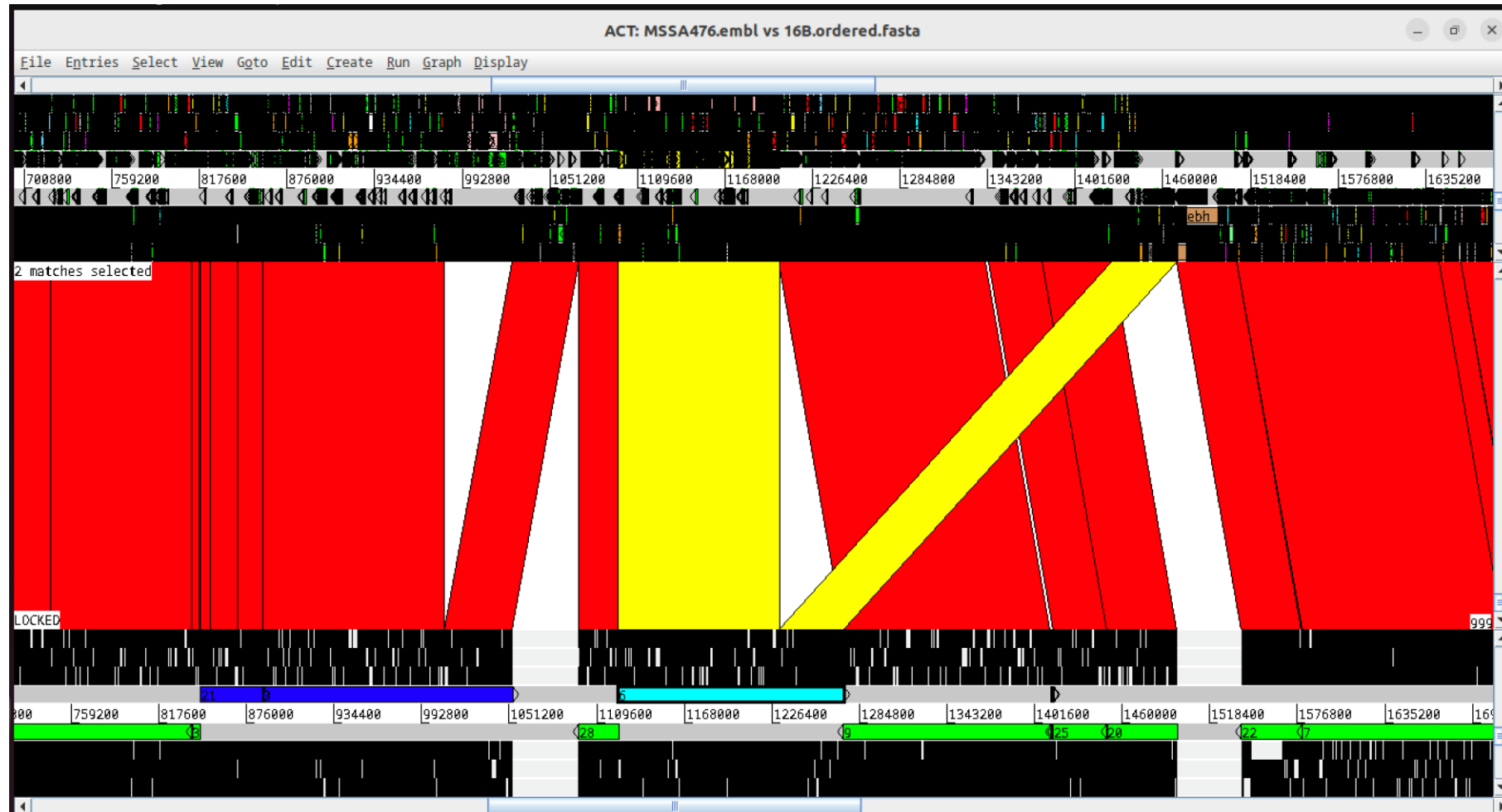
Mismatches

# N's per 100 kbp	0
# N's	0

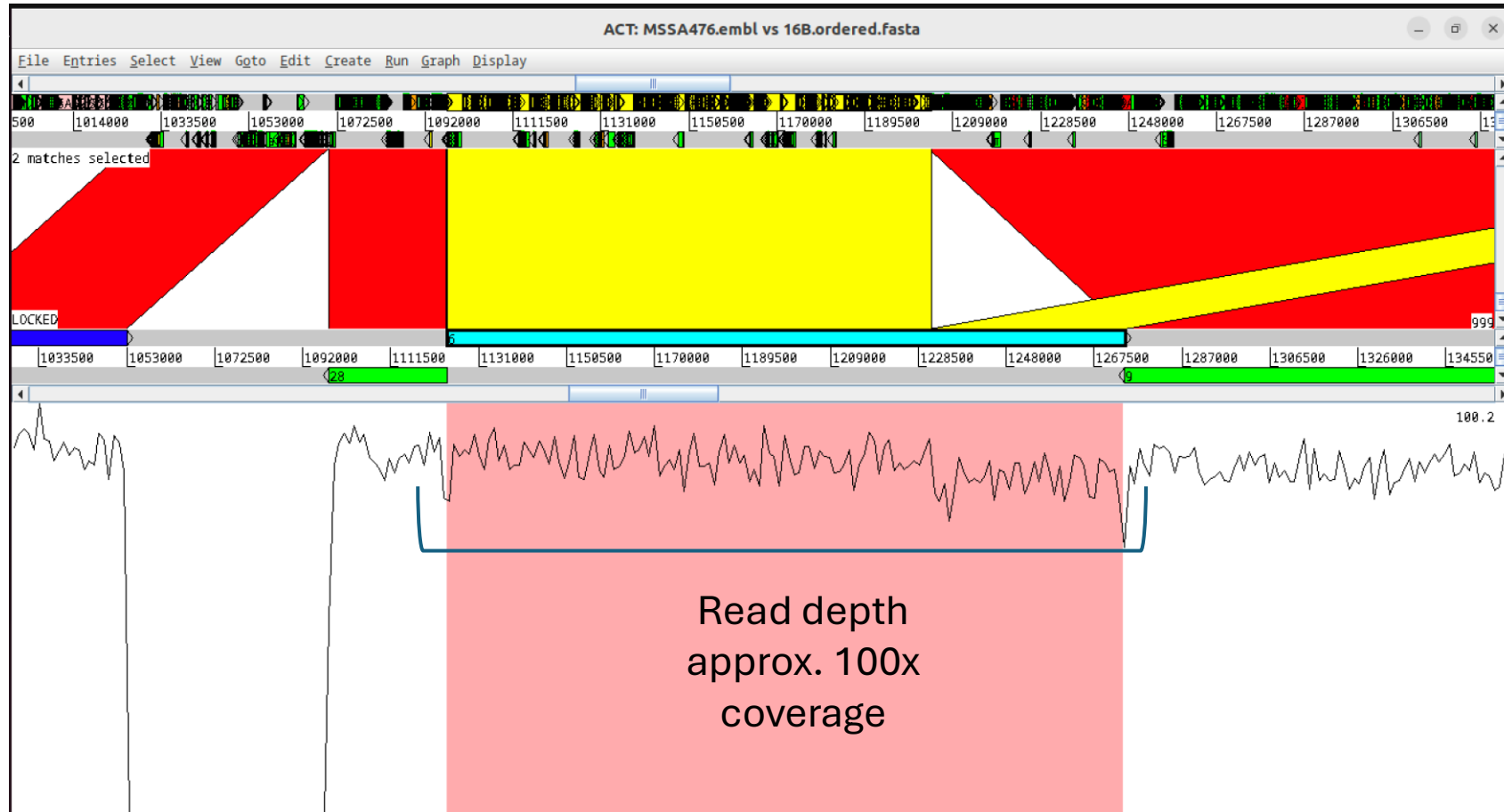
Contig size viewer



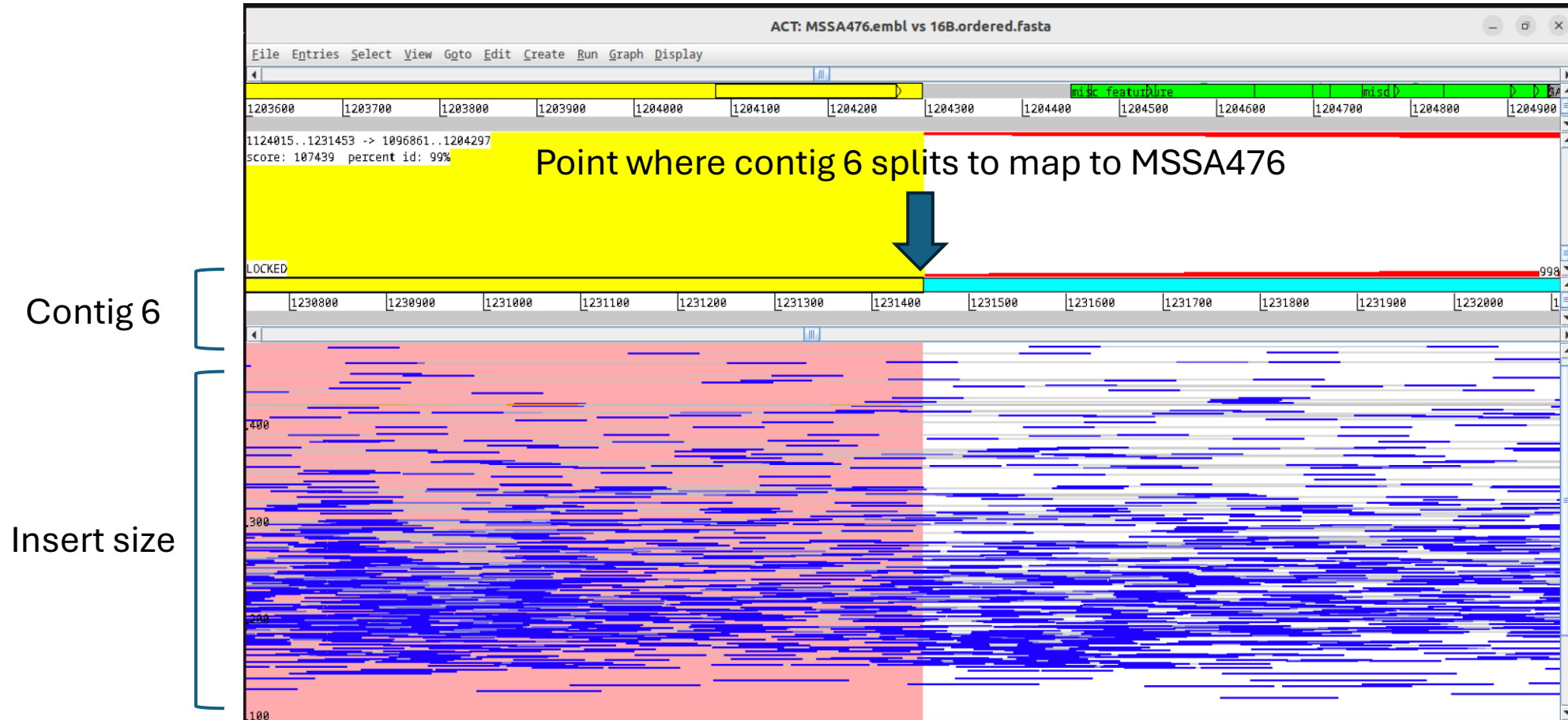
Visualizing blastn matches of reordered assembly



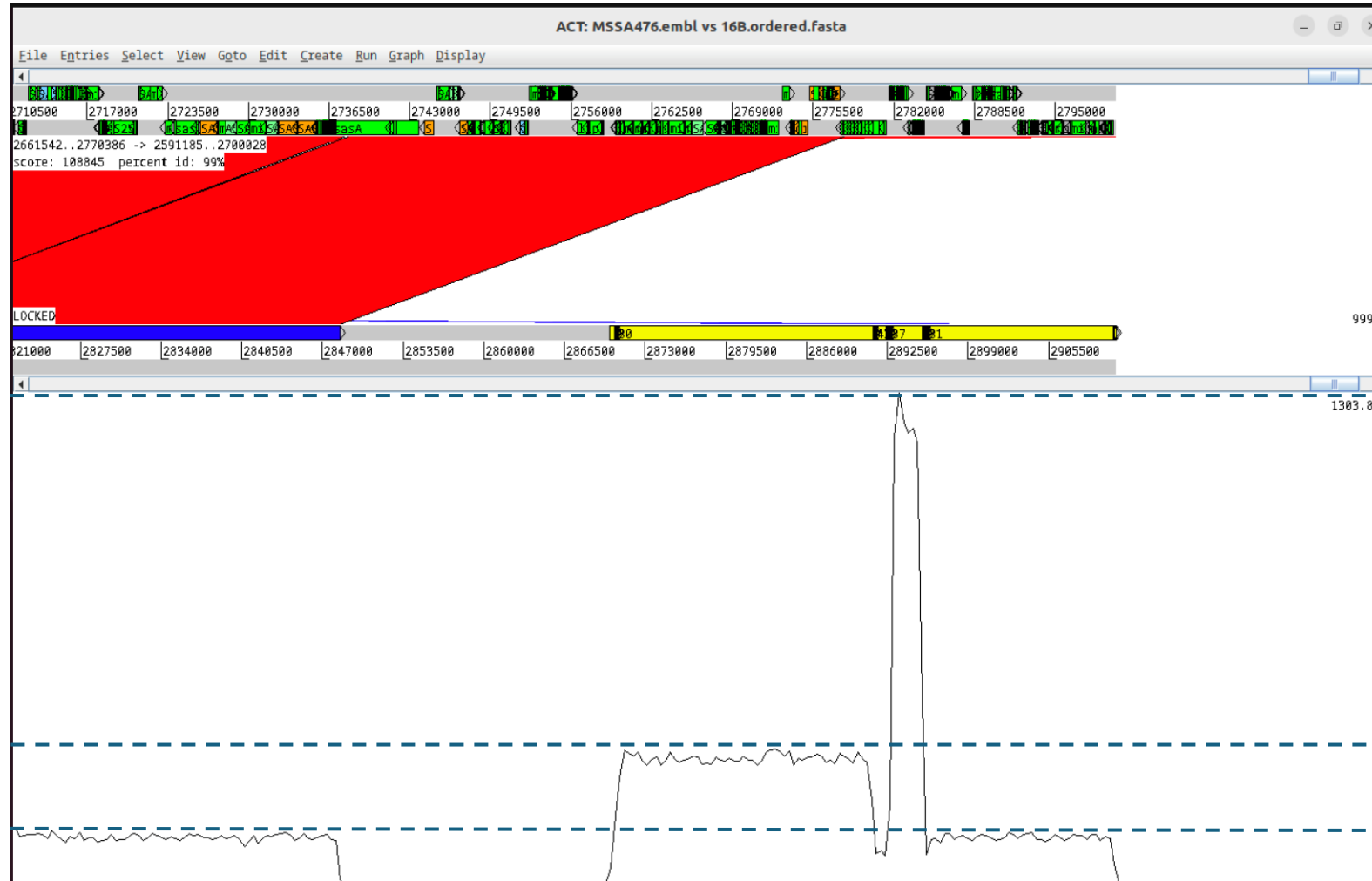
Using mapping data to identify (or rule out) assembly errors



Using mapping data to identify (or rule out) assembly errors



Using mapping data to identify plasmids

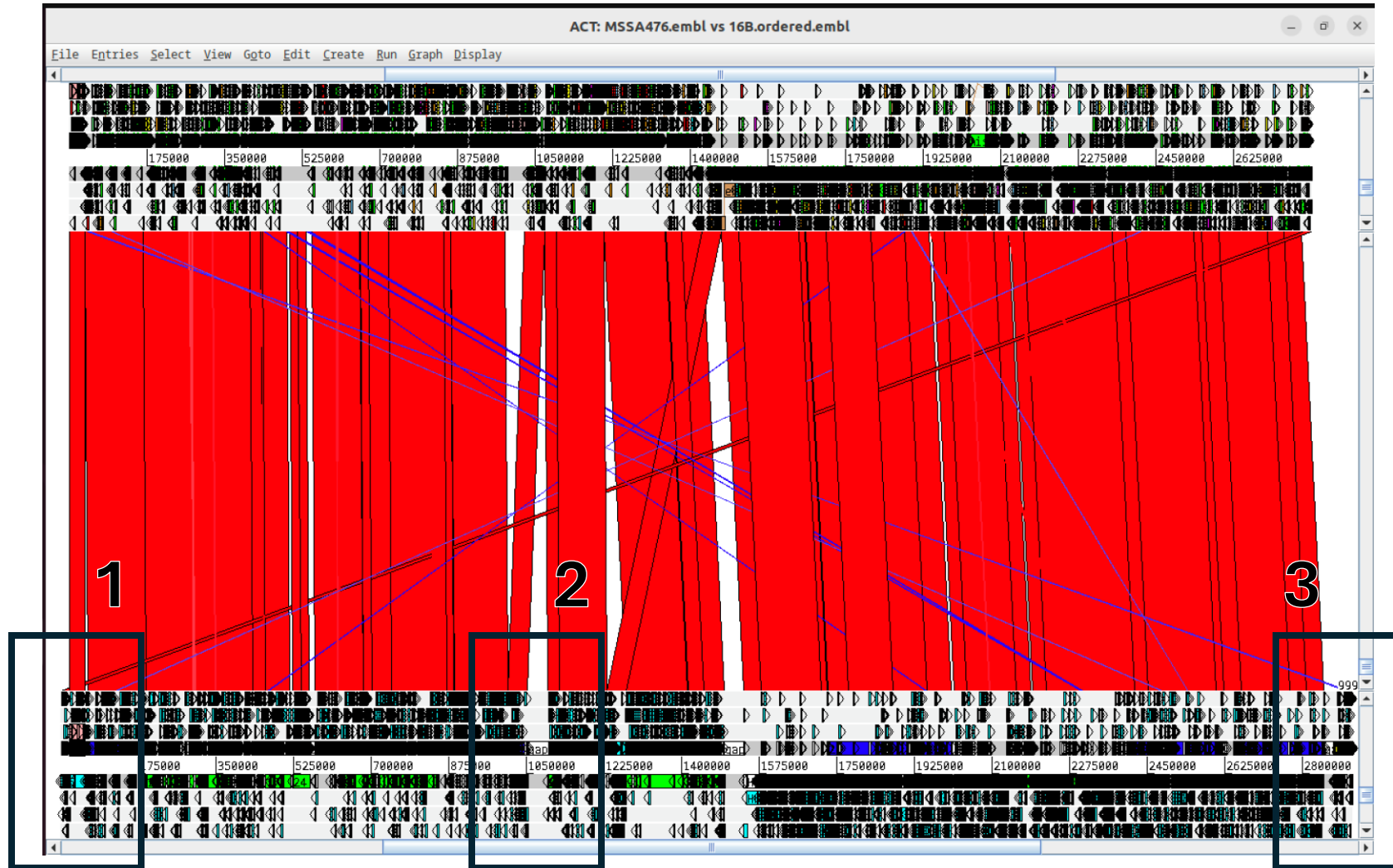


~13x that of
core genome

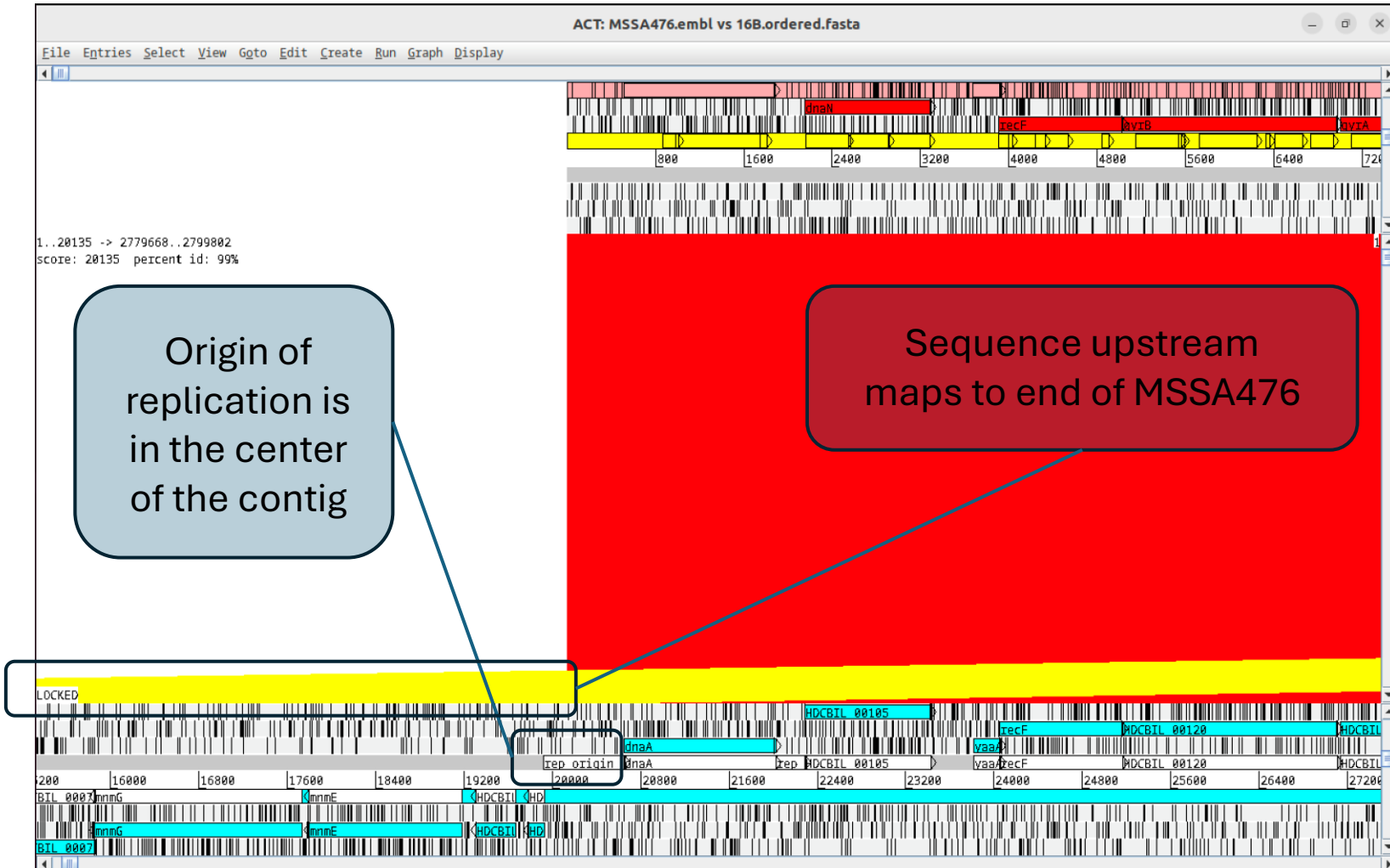
Core
chromosome

~2-3x that of
core genome

Investigating bakta annotations



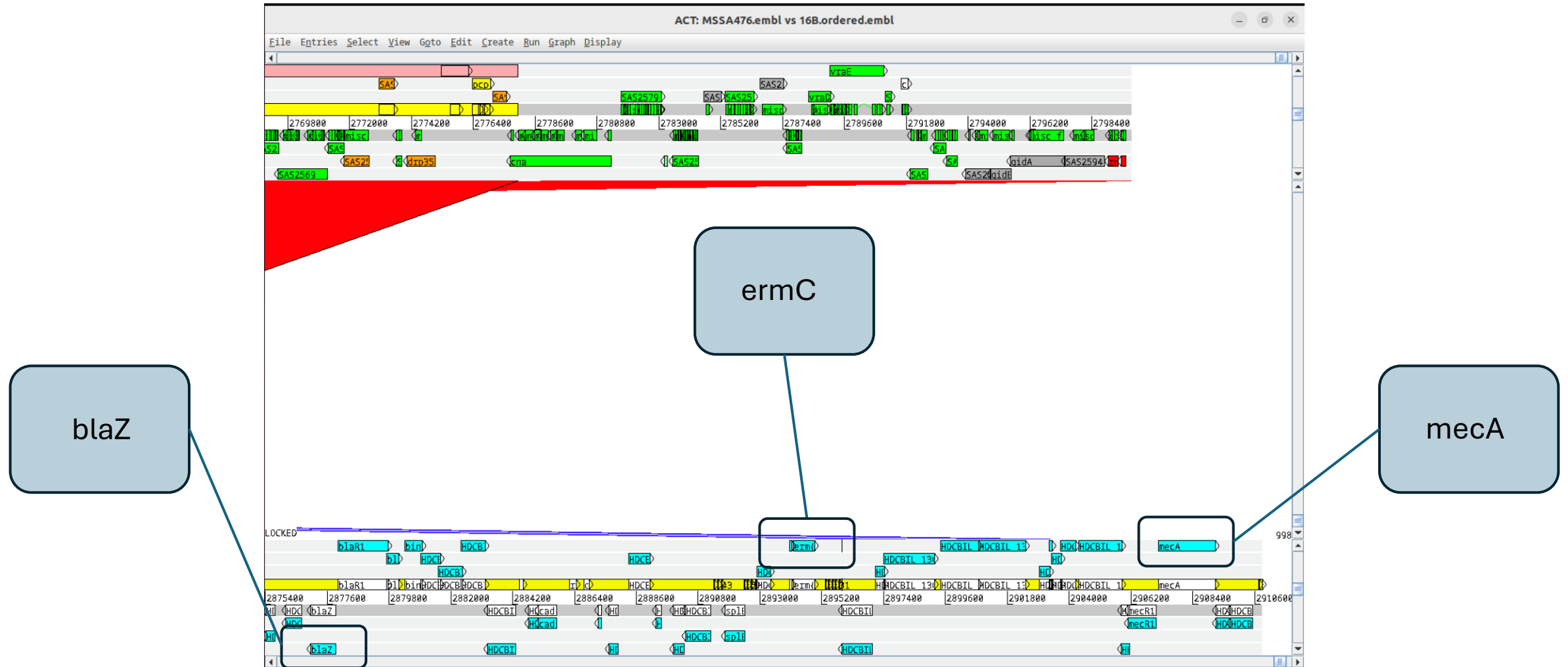
Region 1



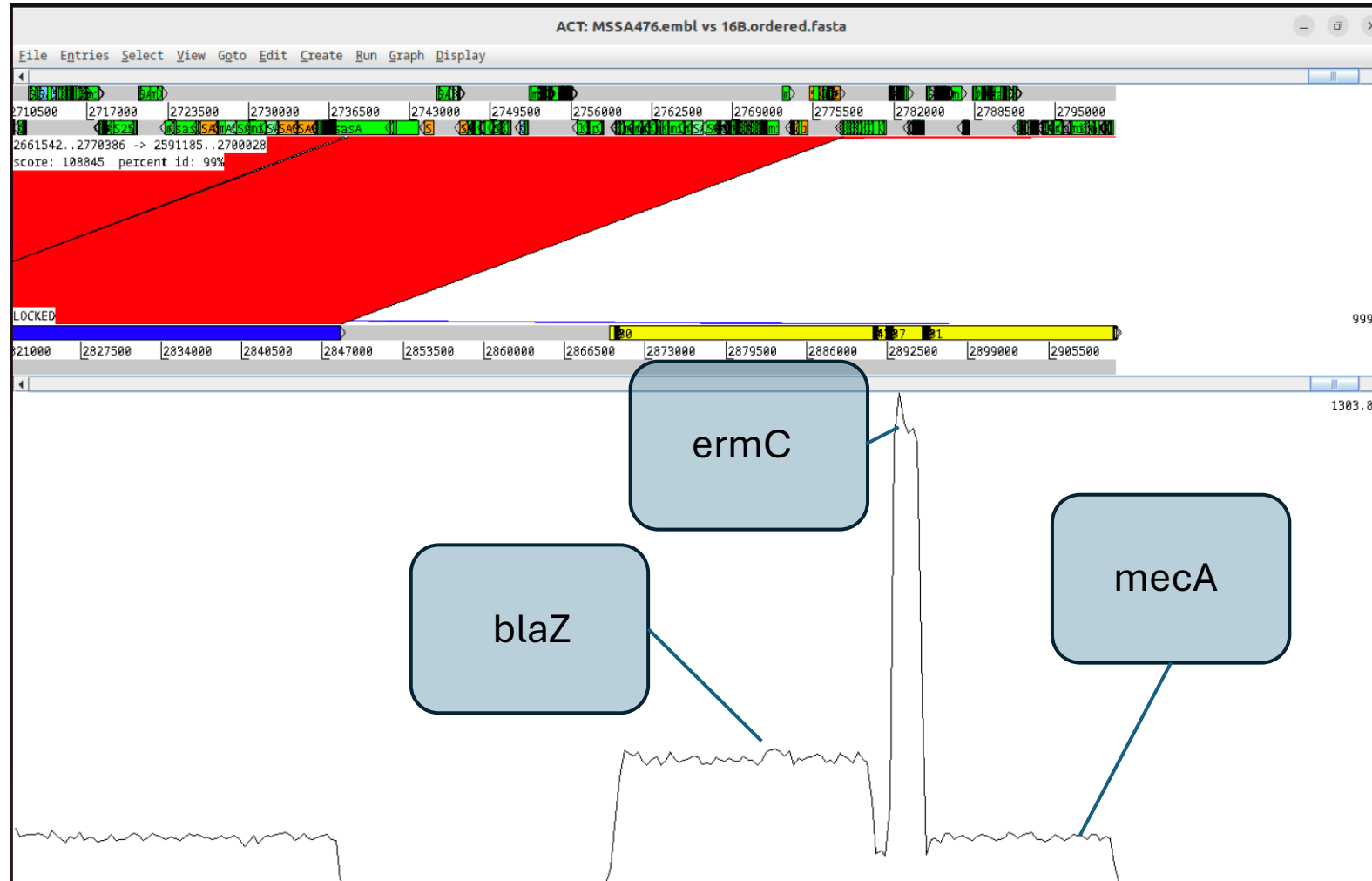
Region 2



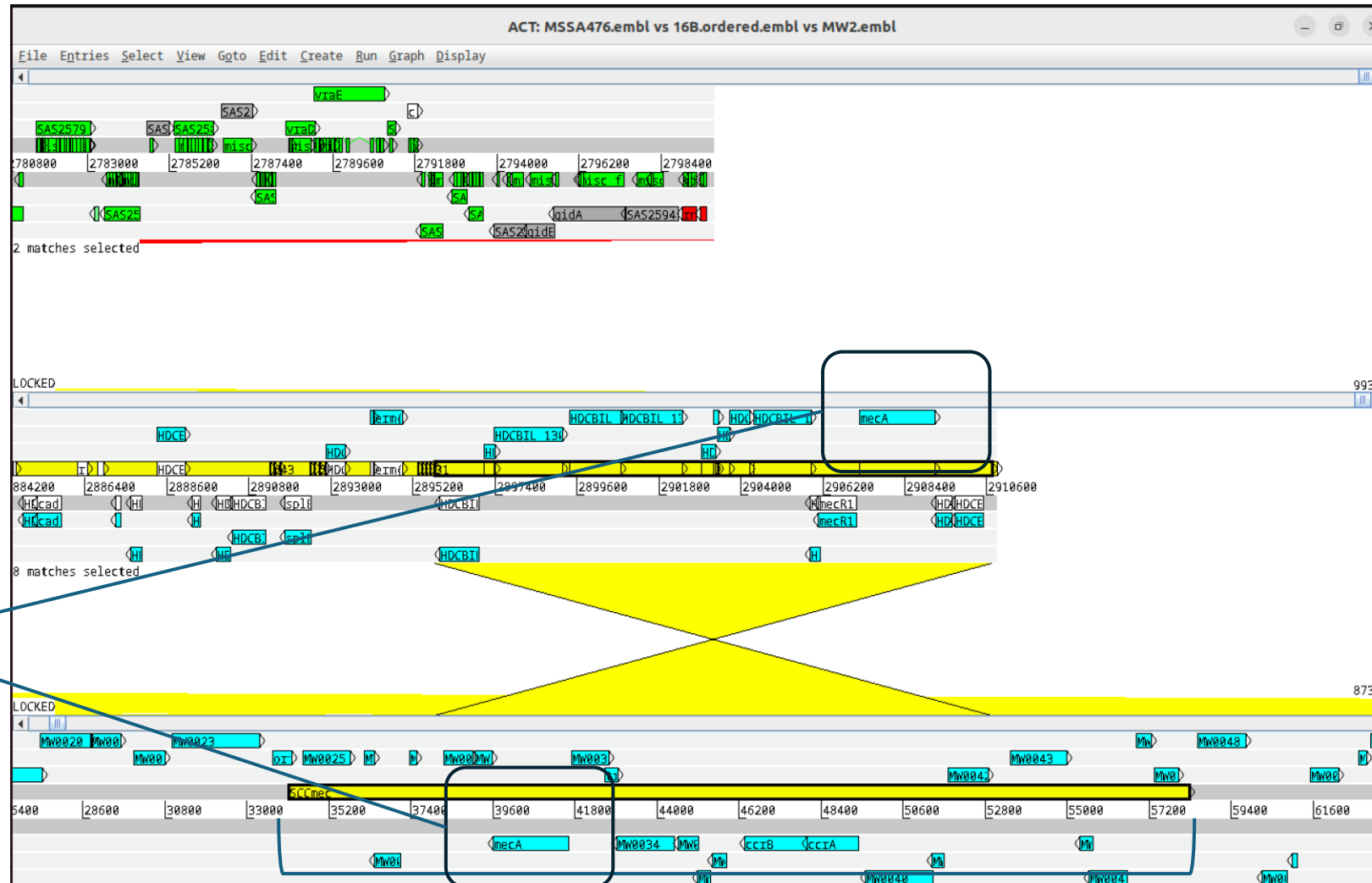
Region 3



Region 3: Using mapping data to identify plasmids



3 way comparison: SCCmec on unmapped contig



Resistance

	blaZ	ermC	fusC	mecA
MSSA476				
16B				
MW2				

Resistance

	blaZ	ermC	fusC	mecA
MSSA476	plasmid		SCCfus	
16B	plasmid	plasmid	SCCfus	SSCmec
MW2	plasmid			SSCmec

SCC (Staphylococcal cassette chromosome)

The research questions

The three ST1 isolates are closely related but exhibit different antibiotic resistance profiles:

- 16B is resistant to penicillin, fusidic acid, methicillin and erythromycin
- MSSA476 is resistant to penicillin and fusidic acid
- MW2 is resistant to penicillin and methicillin

Using a comparative genomic approach we will identify regions of difference, the genetic basis of the antibiotic resistance in 16B, and the genetic mechanisms that drive the evolution of resistance.

Penicillin	<i>blaZ</i>	carried on a plasmid
Fusidic acid	<i>fusC</i>	Staphylococcal cassette chromosome <i>fus</i> element (SCC <i>fus</i>)
Methicillin	<i>mecA</i>	Staphylococcal cassette chromosome <i>mec</i> element (SCC <i>mec</i>)
Erythromycin	<i>ermC</i>	carried on a plasmid

Special acknowledgement to Matt Holden